

Capital One: AI-Enabled Photography and Visual Brand Standards Analysis

Capstone Proposal

Introduction

Companies depend on photography to reinforce their brand across websites, ads, social media, and other digital channels. Some brand elements, like logos, colors, and formatting, are easy to check automatically because they follow clear rules. Photography is harder. Things like lighting, framing, focal point, perspective, depth of field, and tone are qualitative, and two photos can meet the same technical bar while sending completely different messages, one might feel approachable while another feels distant, even if both are “correctly” formatted.

Capital One’s Brand Tech team builds automated tools that check marketing assets against brand standards, but those checks cover things that are easy to measure. Extending that work to photography means dealing with standards that are often loosely defined and inherently subjective.

The problem gets bigger on scale. Photos show up across hundreds of webpages, so before you can evaluate them, you must find them and figure out which images are the photographs you care about. From there, subjective standards need to be turned into criteria that can be applied consistently, and whatever the system decides needs to be explainable to a human reviewer, not just a score with no reasoning behind it.

The core problem is how to evaluate a defined set of photography-related brand standards consistently at scale, when some characteristics can be quantified directly while others require qualitative interpretation. The goal is to develop a system that helps people making brand decisions by producing consistent criterion-level assessments and human-readable explanations, rather than relying only on opaque scores.

Stakeholders

The primary stakeholder is the Capital One Brand Tech team, acting as the project mentor. Their core need is a scalable, explainable system capable of assisting with the consistent evaluation of photography standards that require substantial human interpretation. This tool will help them enforce brand consistency across thousands of digital assets while reducing the bottleneck of manual qualitative reviews.

Internal to Capital One, marketing teams, brand managers, and content creators are key stakeholders. These groups require rapid, actionable feedback on marketing photography. They need the system to provide human readable explanations rather than just arbitrary scores, allowing them to adjust their visual assets efficiently and confidently deploy campaigns.

Externally, the open-source software community and AI researchers represent a significant stakeholder group. Because this project is expected to be open-source and usable without paid software licenses, external developers are also relevant stakeholders. They will benefit from a documented, modular implementation that allows them to adapt the image analysis pipeline for their own unique qualitative assessment challenges.

Photographers and creative professionals are also important stakeholders. They need clear feedback about how characteristics such as framing, lighting, perspective, subject placement, and overall visual tone align with brand expectations. A useful system could help them identify potential issues earlier in the creative process.

User Personas

Patty the Brand Manager



Details: Patty is 35 years old and she's a Brand Manager at Capital One. Her job entails signing off on marketing campaigns made by the marketing team. She has spearheaded several campaigns and would like upcoming campaigns to adhere to a set of decided upon brand standards. She values consistency, efficiency, and clear justification before making decisions.

Goal: She wants a faster, more consistent way to evaluate whether campaign photography aligns with defined brand standards. She would use the system to review criterion-level results and human-readable explanations before approving an image or requesting further review.

Ariel the Content Creator



Details: Ariel is 25 years old and a Content Creator. Her job entails filming content and taking photos for Capital One ad campaigns. Her content has been utilized by several campaigns, and she has catered to various requirements from the marketing team. She prefers feedback early in the creative process rather than the final submission.

Goal: Ariel wants to check whether her photographs align with selected brand standards before submitting them for approval. She would use the product to make sure that all of her content is still in alignment with the overarching branding standards while still meeting the requirements of individual campaigns.

Ben the Marketing Team Lead



Details: Ben is 29 years old and is a Marketing Lead for one of Capital One's marketing teams. He has curated several campaigns for various credit cards. He sets the requirements and messaging for each campaign and directs content creators under him. He values consistency and alignment with campaign goals.

Goal: Ben wants to compare candidate images efficiently and understand how well each one aligns with the selected photography standards. He would use the product to compare previous campaign standards and greenlight content to be used in specific campaigns.

Maya the Brand Tech Engineer



Details: Maya is a 28-year-old software engineer responsible for maintaining and extending brand-technology tools. She has experience with Python, machine learning, and software systems. She values modularity, documentation, and open-source tools that do not depend on paid licenses.

Goal: Maya wants an extensible image-analysis system that can be maintained and improved over time. She would use the modular architecture to test or replace models, add new criteria, and integrate future improvements.

Proposed Solution

To address the challenge of evaluating subjective brand photography at scale, our team proposes building an automated, open-source image analysis pipeline. Rather than relying on a single monolithic model, the solution will utilize a modular architecture that combines traditional computer vision algorithms for characteristics that can be quantified directly, together with pretrained vision or vision language models for characteristics that require broader contextual or qualitative interpretation.

This approach directly solves the core problem of applying loosely defined brand guidelines consistently. By passing images through specialized analysis components, the system will not only evaluate if a photograph aligns with visual standards but will also generate human readable explanations for its decisions. This ensures that marketing teams and reviewers receive actionable feedback rather than opaque scores. The system is intended to support, rather than replace, human brand judgement. Its output will provide evidence and

explanations that users can consider when deciding whether an image conforms to the selected standards.

While Capital One acts as our mentor to provide industry requirements and context, the direction and implementation of this project remain entirely student driven. The team will maintain full ownership over the system design, model selection, and evaluation methods. The team will evaluate existing open-source computer vision and vision-language models and may develop custom preprocessing, prompting, scoring, or integration components where appropriate rather than developing every model from scratch. The prototype will rely on open-source software components and models that can be used without paid licenses, supporting reproducibility, and future use by the mentor team. For the one-semester capstone, the final deliverable will be a lightweight software prototype that allows users to upload marketing assets and receive structured, interpretable evaluations, successfully bridging the gap between subjective brand standards and automated content assessment.

Related Work / Competition

Adobe GenStudio for Performance Marketing and Frontify are established platforms for maintaining brand consistency (Adobe, 2026; Frontify, n.d.). Adobe GenStudio validates generated marketing content against uploaded brand guidelines, produces compliance scores, identifies potential issues, and supports approval workflows (Adobe, 2026). Frontify centralizes brand guidelines and approved digital assets while making those guidelines accessible to users and AI systems (Frontify, n.d.). These platforms demonstrate the value of automated brand governance, but their documented capabilities emphasize asset management, content generation, guideline access, and workflow compliance. Our prototype instead focuses specifically on explainable, criterion-level evaluation of existing photography using open-source components.

Traditional computer-vision libraries such as OpenCV provide measurable image-processing capabilities, including color conversion, filtering, geometric transformations, contours, and histograms (OpenCV, n.d.). These techniques can support objective measurements such as color distribution, brightness, contrast, and subject positioning, but these techniques are well suited to measurable image properties, but on their own they are limited in assessing broader semantic or qualitative brand characteristics.

Image-aesthetic assessment research addresses more subjective characteristics. Neural Image Assessment (NIMA) predicts distributions of human ratings rather than only a single average aesthetic score (Talebi & Milanfar, 2018). More recent work, such as Vision-Language Aesthetics (VILA), uses vision-language learning to generate richer aesthetic descriptions (Ke et al., 2023), while personalized image-aesthetic assessment research

demonstrates that aesthetic judgments vary among viewers (Yang et al., 2022). Our prototype combines these directions: traditional measurements for criteria that can be quantified reliably, multimodal analysis for criteria requiring broader contextual interpretation, and human-readable explanations tied to a mentor-defined photography rubric.

MVP

The minimum viable product will include following core functionality:

- Conduct an initial research and literature review to identify a manageable subset of photography standards suitable for automated evaluation.
- Define approximately 4-6 target photography criteria for the MVP, based on the research findings and mentor guidance.
- Accept one or more photographs as input for evaluation.
- Evaluate each image against the selected photography criteria using appropriate image-analysis methods.
- Investigate and compare multiple potential approaches that may include traditional computer vision techniques, pretrained vision models, prompt-based evaluation, etc.
- Generate criterion-level assessments for each selected photography standard.
- Provide an overall summary indicating how well the photograph aligns with the defined photography standards.
- Provide human-readable explanations for the generated assessment so that users can understand the reasoning behind the result.
- Present results through a lightweight software interface or structured reporting workflow.
- Maintain a modular codebase so that individual analysis methods or AI models can be replaced, extended, or improved.
- Use open-source components that do not require paid licenses.
- Include documentation describing how to run and use the system.

Stretch Goals

- Automatically identify and extract relevant photographs from public webpages for analysis.
- Expand the number of supported photography standards beyond the initial MVP subset.
- Perform site-wide or large-batch photography analysis.

- Compare photography characteristics across competitor websites for benchmarking
- Track visual-standard changes over time (historical tracking) for the same website to catalogue variances.
- Suggest possible corrective actions in addition to scores and explanations.
- Extend the system to analyze photography standards in video from ad campaigns.
- If results demonstrate sufficient novelty, conduct publication-oriented experiments, and prepare a research manuscript.

Workflows

User Workflow

The primary workflow begins when a user provides one or more photographs for evaluation. The system first validates and preprocesses the input images, so they are in a consistent format for analysis. The user then selects from the set of photography standards. Each image is passed through the analysis pipeline, where the system evaluates selected visual characteristics such as lighting, framing and composition, focal point or subject placement, perspective, color characteristics, and selected qualitative attributes such as tone or visual impression. Different criteria may be evaluated using different computer vision or multimodal AI components. These results are then translated into a human-readable explanation.

The system combines the criterion-level results into a structured assessment. For each criterion, the user receives an evaluation along with a human-readable explanation describing the factors that influence the result. The interface also provides an overall summary of how well the photograph aligns with the selected photography standards.

For multiple-image input, the same process is repeated consistently across all images, allowing the user to compare results. The user can then review the assessments and explanation and use them as decision support when determining whether an image requires further review or adjustment.

Development and Evaluation workflow

The team first identifies the target photography criteria through literature and standards review. Candidate methods are then selected for each criterion and tested on representative images. The outputs are compared for consistency, interpretability, and usefulness. The most suitable methods are integrated into the modular pipeline and then tested through the software interface before the final documentation and release. The proposed user and development workflows are summarized in Figure 1.

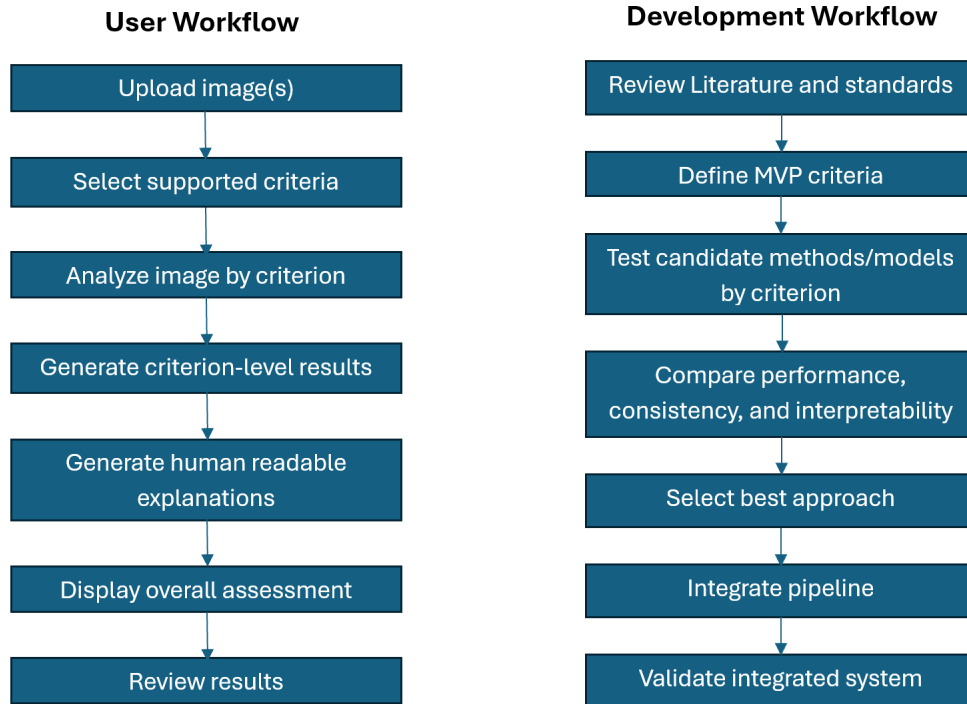


Figure 1. Proposed user evaluation and development/validation workflows for the photography standards analysis system.

Implementation

The proposed solution will be implemented as a lightweight software application built around a modular image-analysis pipeline that combines computer vision and AI techniques. Users will upload one or more photographs through a graphical interface. Input photographs will first be standardized and then passed through individual analysis components responsible for evaluating selected photography characteristics.

Different photography criteria may require different evaluation approaches. Characteristics that can be quantified reliably, such as color distribution, may be analyzed using traditional image processing and computer vision techniques, while criteria requiring broader interpretation may be evaluated using more advanced techniques like pretrained models or vision-language models.

The team will not assume that a single algorithm is appropriate for every photography standard. After reviewing relevant literature, multiple candidate analysis methods will be identified and tested separately for the selected MVP criteria using representative images and defined evaluation criteria. Based on the experiments, the most suitable methods will be integrated into the final modular pipeline. The system will generate structured criterion-

level assessment, an overall summary, and human-readable explanation supporting the results.

The software interface will present uploaded images alongside their evaluation results and explanations in a clear format. The modular architecture will allow individual analysis methods, models, or brand-standard definitions to be replaced or extended as the project evolves.